

UTKARSH GUPTA

Backend & AI Engineer | Python · FastAPI · LLMs · AWS

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Backend & AI Engineer with 4+ years building production-grade systems for enterprise clients including PwC and Aditya Birla Group. Specializes in LLM-powered applications, FastAPI microservices, and cloud-native deployments on AWS and Azure. Proven track record of delivering GenAI pipelines — from RAG and document intelligence to risk prediction — alongside robust financial API integrations at scale. Experienced across the full delivery lifecycle: design, CI/CD, serverless deployment, and SRE.

EXPERIENCE

NeoSoft Technologies – Mumbai, India *Dec 2025 – Present*

Python Developer (Client: Aditya Birla Group)

- Built and maintained third-party API integrations for financial workflows serving Aditya Birla Group's 10M+ customer base
- Integrated Razorpay, Digio, and Tartan APIs covering:
 - PAN verification & PAN linkage validation
 - E-mandate setup & verification
 - Payment processing and transaction flows
- Designed secure REST APIs using FastAPI/Django; reduced integration failure rate by ~30% via structured retry and error-handling layers
- Enforced compliance and security standards (JWT, OAuth, field-level encryption) for sensitive financial data
- Implemented data validation and transformation layers ensuring clean downstream consumption of third-party responses
- Collaborated with cross-functional teams across audit, QA, and DevOps to debug, test, and ship integrations on schedule

Mobifly – Gurugram, Haryana *Mar 2024 – Dec 2025*

Python Developer

PWC E-invoice Solution (Microservices | Server & Serverless)

- Developed server and serverless services handling IRN (Invoice Reference Number) and E-Way Bill APIs for 1000+ daily transactions
- Built automated email alert service reducing manual follow-up effort by ~40% for clients and internal team
- Implemented data transformation pipeline supporting 5+ client-specific formats with zero data-loss guarantees
- Drove SRE practices (error budgets, alerting, runbooks) achieving 99.5%+ service availability
- Managed multi-cloud deployments across Azure, AWS, and ADO with zero-downtime release cycles

PWC Navigate Tax Hub (Microservices | Server & Serverless)

- Architected and deployed FastAPI + PostgreSQL backend on AWS Lambda (serverless), cutting infrastructure cost by ~35% vs. EC2
- Built GenAI pipeline using OpenAI + LangChain for tax document Q&A, summarization, and risk prediction — processing 500+ documents/month
- Designed multi-format file processing system supporting PDF, Excel, and scanned documents via OCR (Tesseract, Docling, AWS Textract)
- Automated CI/CD via Azure DevOps, reducing deployment time from 2 hours to under 15 minutes
- Implemented Redis caching layer improving average API response time by ~60%
- Exported structured AI-extracted data to Excel reports consumed directly by PwC tax analysts

Empowered Wings (OPC) Pvt. Ltd *Jul 2022 – Mar 2024*

Python Developer

- Designed and implemented a scalable email infrastructure (AWS SNS + SQS) capable of handling 50K+ messages/day with guaranteed delivery
- Built and optimized 15+ RESTful APIs using Django and FastAPI; improved average response time by 25% through query optimization
- Integrated OAuth and JWT authentication supporting role-based access for multi-tenant application
- Leveraged Django ORM and SQLAlchemy for complex query optimization, reducing DB load by ~20% on high-traffic endpoints
- Deployed and maintained applications on AWS and Azure with 99%+ uptime across production environments
- Implemented Celery + Redis task queue for async background jobs, offloading 30%+ of synchronous processing load
- Set up CI/CD pipelines via Azure DevOps and GitHub Actions, enabling daily automated deployments with rollback support

SKILLS

AI & GenAI: LangChain, OpenAI APIs, LLM Orchestration, RAG Pipelines, Vector Databases (ChromaDB), Prompt Engineering, LLM-as-Judge

OCR & Document AI: Tesseract, Docling OCR, AWS Textract, Multi-format file processing (PDF, Excel, scanned docs)

Backend & APIs: Python, FastAPI, Django, Flask, DRF, RESTful APIs, Microservices, Serverless (AWS Lambda, Azure Functions)

Databases: PostgreSQL, MySQL, Redis (caching), Firebase, SQLAlchemy, Django ORM

Cloud & DevOps: AWS (EC2, S3, Lambda, SNS, SQS), Azure (DevOps, ADO, Functions), Docker, Kubernetes, CI/CD

Auth & Security: OAuth, JWT, RBAC, Field-level Encryption, Compliance (financial data)

Performance & Async: Celery, Redis, Background Jobs, Task Scheduling, SRE

Frontend: HTML, CSS, JavaScript, Bootstrap, React.js

Version Control: Git, GitHub, Azure DevOps (ADO)

CERTIFICATIONS

Anthropic Academy — AI Fluency Certification 2025

Foundations of LLMs, prompt engineering, responsible AI, and practical Claude API usage

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University, Lucknow 2016 – 2020

Bachelor of Technology | CGPA: 8.03

Intermediate

High School